Seminar in
Laboratory of Visual Intelligence and Pattern Analysis (VIPA)

A Review On Adversarial Attacks

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College of Computer Science and Technology
Zhejiang University
4/6/2018, Hangzhou

Outline

- Introduction
- Basic Concepts
- Adversarial Attacks on DNN
- Discussions

Introduction

- **Adversarial Attack**
  - Intentionally manipulate data to **cheat** machine learning systems.

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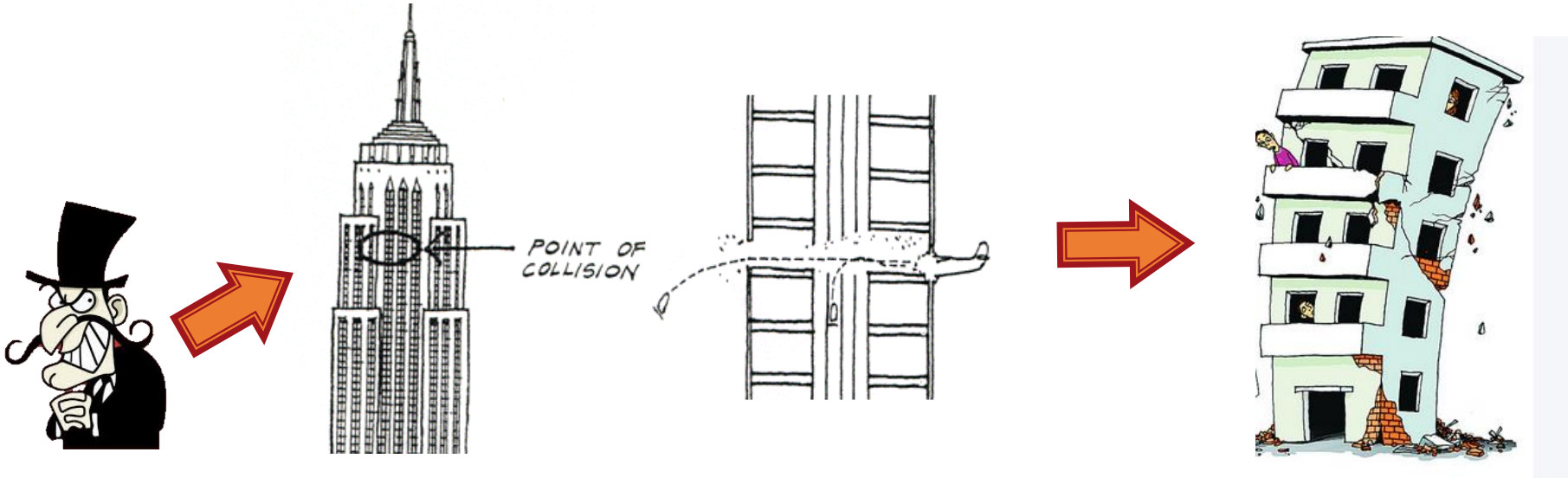
- **Adversarial Learning**
  - An **independent** research area to develop **Adversarial-Aware** machine learning systems in **Adversarial Environment**.
  - **A Game** between two players: the **Classifier** and the **Attacker**

Adversarial Environment:



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  - Intentionally manipulate data to **cheat** machine learning systems.
- **Adversarial Learning**
  - An **independent** research area to develop **Adversarial-Aware** machine learning systems in **Adversarial Environment**.
  - **A Game** between two players: the **Classifier** and the **Attacker**
  - Adversarial Learning for **Big Data**, Adversarial Learning for **Computer Vision**, …

TKDE

TPAMI, TIFS [Information Security Top Journal]

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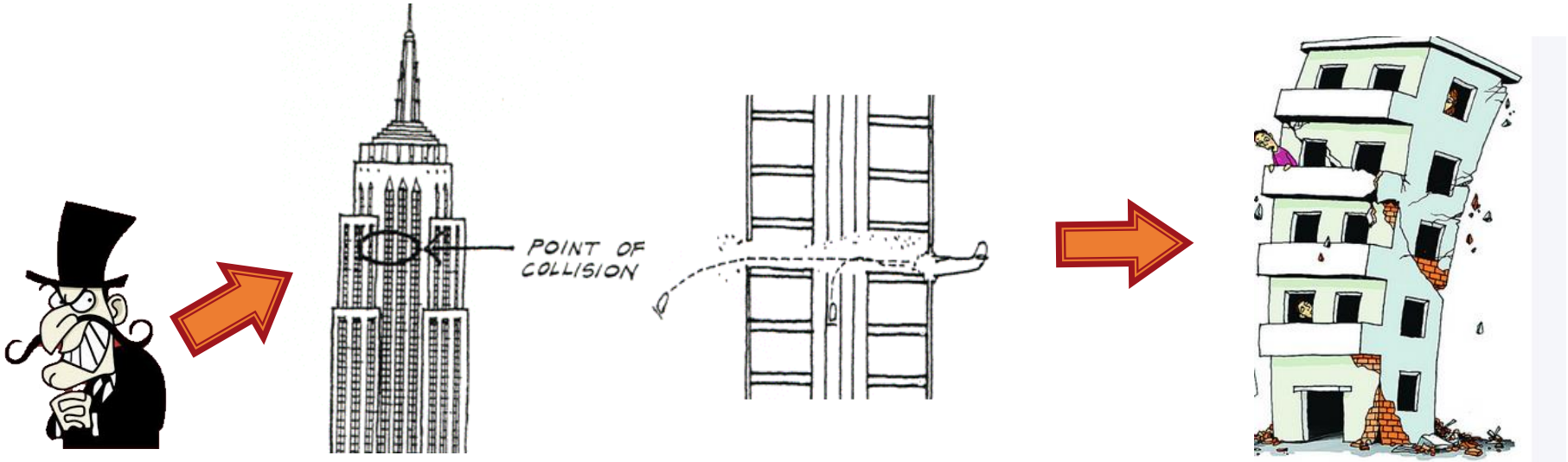
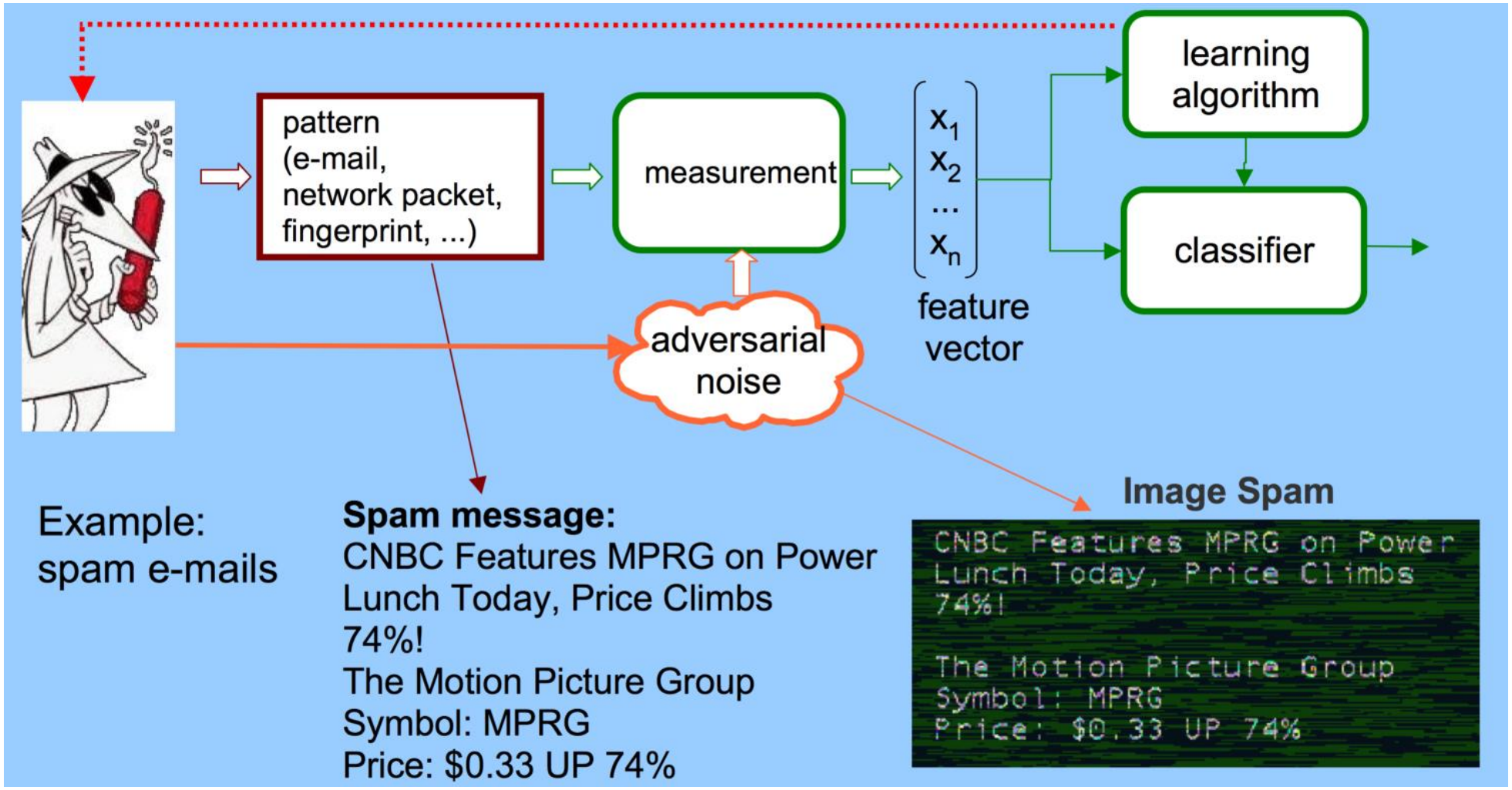


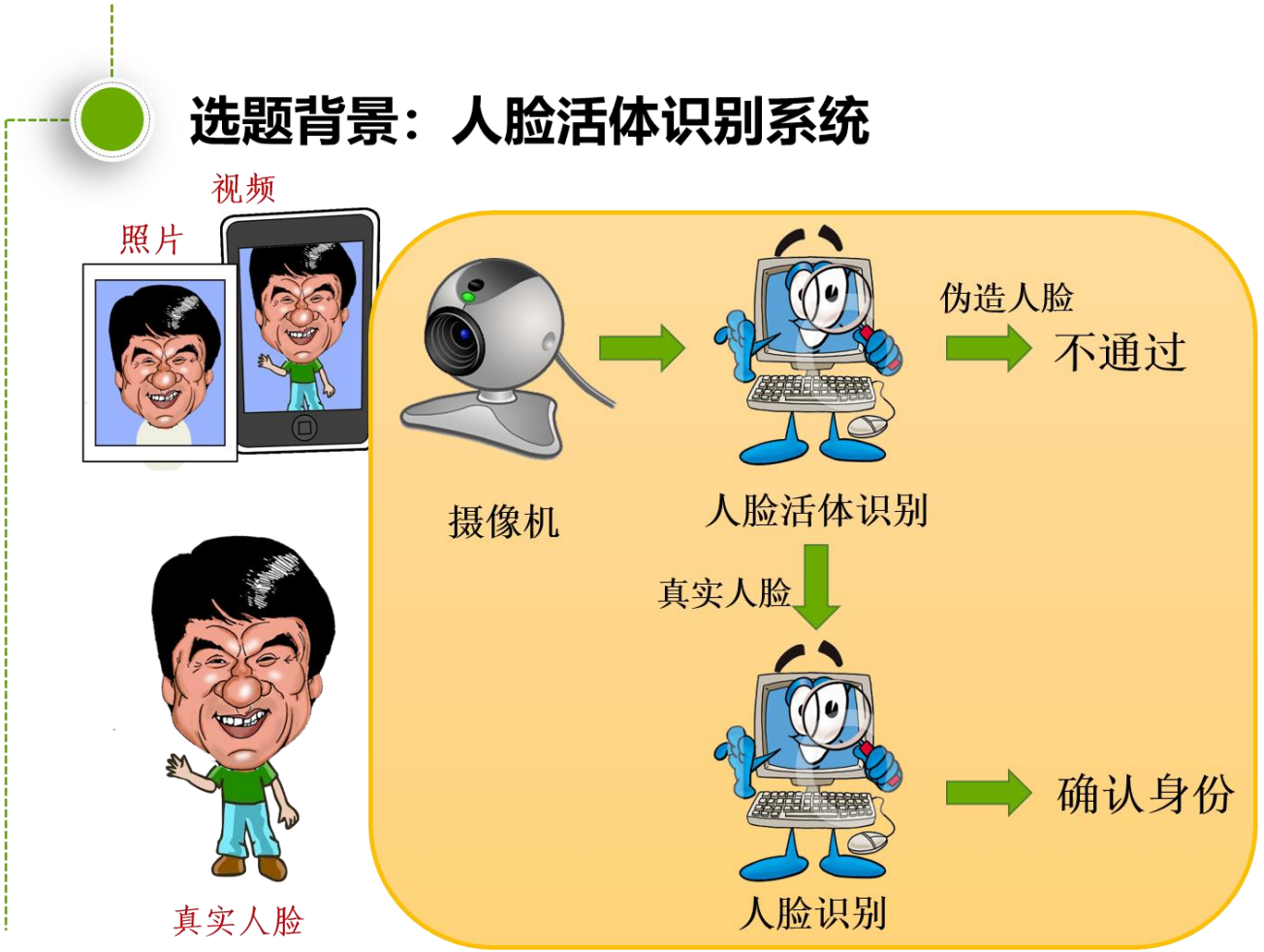
Introduction

- Example Applications: **Spam E-mails Filtering**



Introduction

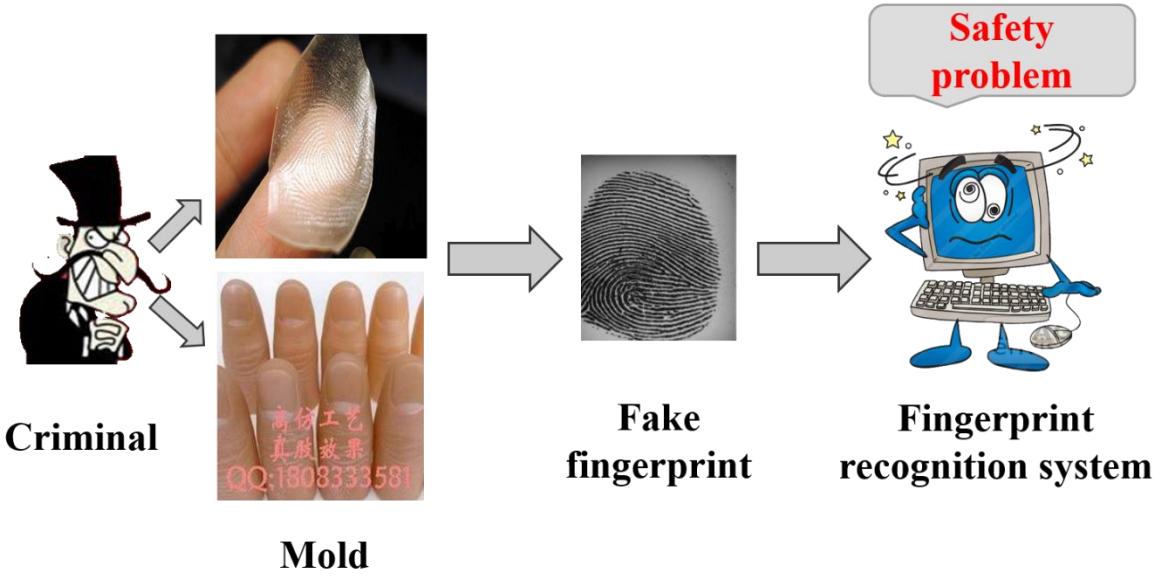
- Example Applications: **Face Liveness Detection**

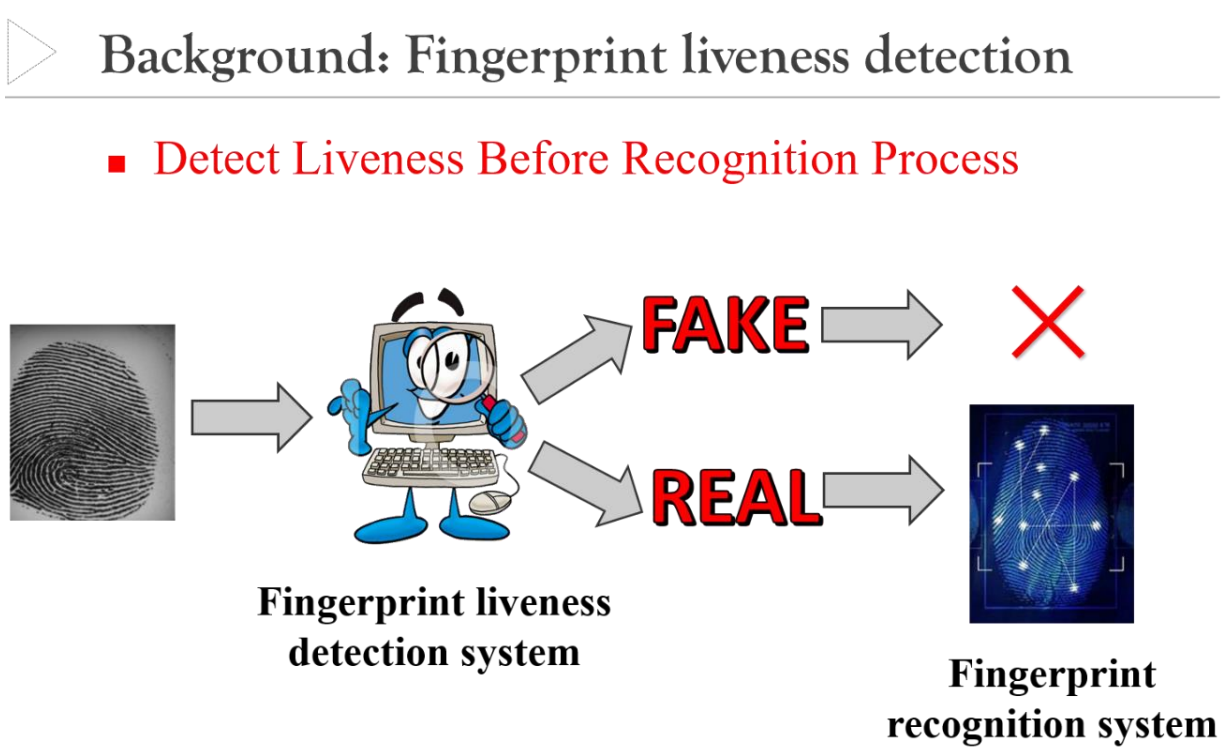


Chan, P.P., Liu, W., Chen, D., Yeung, D.S., Zhang, F., Wang, X. and Hsu, C.C., 2018. **Face Liveness Detection Using a Flash Against 2D Spoofing Attack**. *IEEE Transactions on Information Forensics and Security*, *13*(2), pp.521-534. (TIFS, CCF A)

Introduction

- Example Applications: **Fingerprint Liveness Detection** **[LivDet Competition]**





Motion Blur Attack against Fingerprint Liveness Detection and its Countermeasure

Introduction: Pre-neural Era

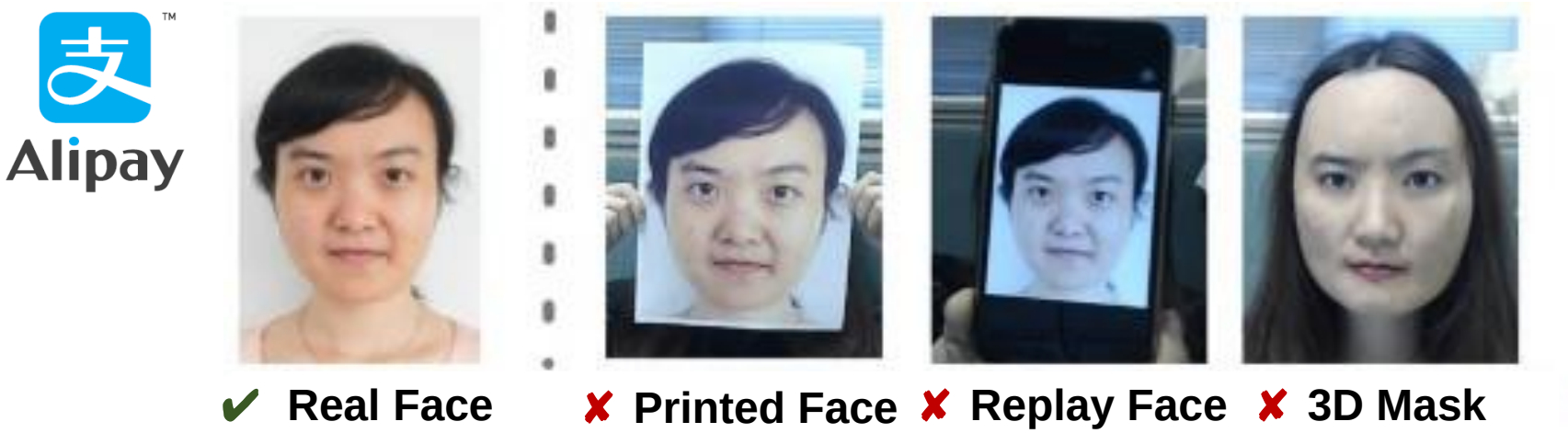
- **Adversarial Learning** is **NOT** a novel concept emerging in Neural Era[1].
  - Previously, **Adversarial Learning** is built on the basis of **traditional machine learning system**.
  - To my knowledge, even some researchers only focus on Adversarial Learning related projects years ago[2].
  - Why not so popular **in the pre-neural era**?
    - **Limited application** of Adversarial Learning in practice in the pre-neural era.
    - The targeted machine learning system needs to be **popular** first.

1. Lowd D, Meek C. Adversarial learning. In Proceedings of the eleventh ACM SIGKDD international conference on Knowledge discovery in data mining 2005 Aug 21 (pp. 641-647). ACM.

2. Nelson, Blaine Alan. Phd Thesis: *Behavior of Machine Learning Algorithms in Adversarial Environments*. University of California, Berkeley, 2010.

Introduction: Neural Era

- **Adversarial Learning** is becoming a trending topic **[11 papers in ICLR 2018].**
  - Lots of CNN-based algorithms come into every field in people' lives, e.g. **face recognition and fingerprint recognition** in our lab.
  - The security issue is then critical, especially those for **financial usage**.
  - e.g. Face recognition for Alipay.



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- **Patterns** in the papers on Adversarial Learning:
  - **1) As an attacker:** Propose some **evasion attack** techniques.

    (maybe it's enough for an pioneering work to just propose evasion attack techniques)
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Our Focus Today

Basic Concepts

- **Adversarial Sample / Adversarial Example**
  - Examples that are **similar** to examples in the true distribution, but that **fool** a classifier.
  - i.e., **modified** versions of a clean image that are **intentionally perturbed** (by adding noise) to confuse a machine learning system.

“扰动”

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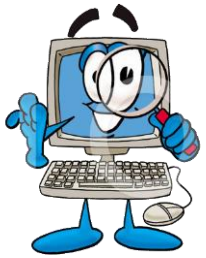

Classifier:
84% for sure it is a **Fat Panda !**

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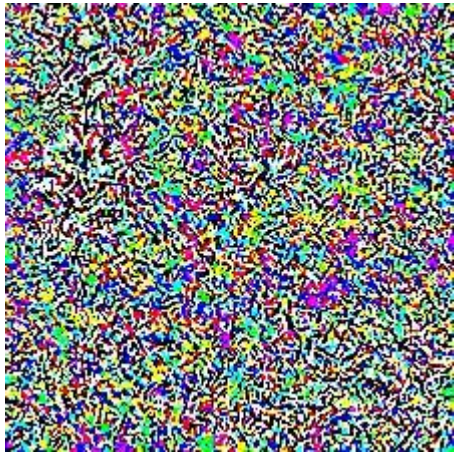

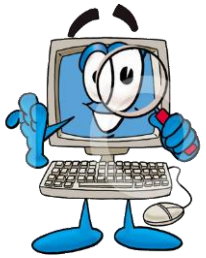

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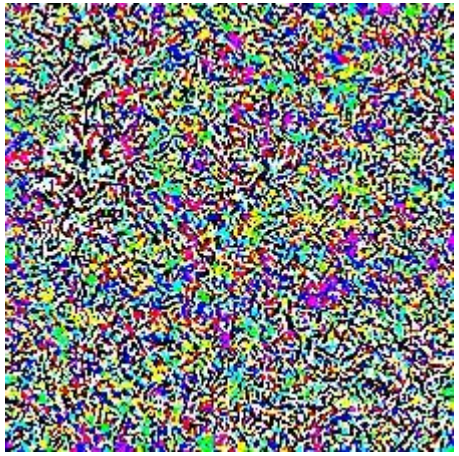

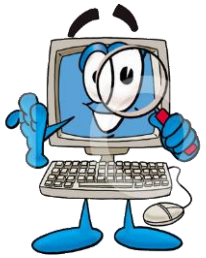

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Adversarial Perturbation (Noise)

67%: Maybe Capuchin ?

Basic Concepts

- **Black-box Attacks**
  - The adversary has **no knowledge** of the target model.
- **Semi-black-box Attacks**
  - The adversary has a **limited knowledge** of the model (e.g. architecture) but definitely does not know the **parameters**.
- **White-box Attacks**
  - The adversary has a **complete knowledge** of the target model.

Adversarial Attacks: A Taxonomy

Adversarial Attacks

- Generative model
- Classification model
  - **White-box Attacks**
    - ① Box-constrained L-BFGS
    - ② Fast Gradient Sign Method (FGSM)
    - ③ Iterative Fast Gradient Sign Method (Iterative FGSM)
    - ④ One Pixel Attack
    - ⑤ Universal Adversarial Perturbations
  - **Black-box Attacks**
    - Antagonistic Network for Generating Rogue Images (ANGRI)
    - Universal Perturbations for Steering to Exact Targets (UPSET)
- Reinforcement Learning
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Adversarial Attacks: Details

- ① **Box-constrained L-BFGS**
- Problem formulation:
  - 1. Assume that we have a network $\boldsymbol{f}$ parametrized by $\boldsymbol{\theta}$ that maps a sample $\boldsymbol{x}$ to **real label** $\boldsymbol{y}$.

Szegedy, C., Zaremba, W., Sutskever, I., Bruna, J., Erhan, D., Goodfellow, I., & Fergus, R. (2013). Intriguing properties of neural networks. In ICLR.

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  - 3. Formulation:

$$\arg\min \|\underbrace{\rho}_{\text{perturbation}}\|_2 \text{, subject to } f(x+\rho;\theta) = \underbrace{y_t}_{\text{target label}}$$

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perturbation target label

For optimization

$$\boxed{\arg\min_{\rho} c|\rho| + \mathcal{L}(f(x+\rho;\theta), y_t)}$$

positive scalar loss function

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Adversarial Attacks: Details

- ① **Box-constrained L-BFGS**
- Discussions:
  - 1. **Multi-step** procedure
  - 2. **Target** attack (y_t)
  - 3. **Slow**. Several minutes for a single attack.
  - ★ 4. Perturbations computed for **one network** can also fool **other networks**. This ability is called **"Transferability"** in this field.

For optimization →

$$\arg\min_{\rho} c|\rho| + \mathcal{L}(f(x+\rho;\theta), y_t)$$

positive scalar loss function

Formal definition: The ability of an adversary example to remain effective even for the models **other than** the one used to generate it.

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Adversarial Attacks: Details

- ② **Fast Gradient Sign Method (FGSM)**
  - **Single-step** and **Fast** Attack. Based on **Linear Perturbation**.
  - Formulation:

$$\rho = \epsilon \cdot \mathrm{sign}(\nabla_x \mathcal{L}(f(x;\theta), y_t))$$

perturbation

small scalar (e.g., 0.1, 0.007)

FGSM hypothesizes that the designs of modern DNN intentionally encourage **linear behavior** for computational gains.

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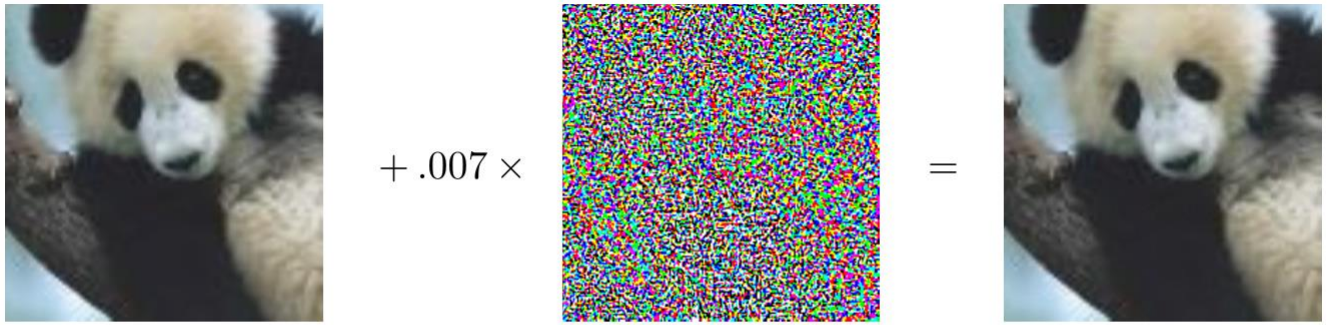
e.g.



$\boldsymbol{x}$
"panda"
57.7% confidence

$+ .007 \times$

$\mathrm{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$
"nematode"
8.2% confidence

$=$

$\boldsymbol{x} + \epsilon\mathrm{sign}(\nabla_{\boldsymbol{x}} J(\boldsymbol{\theta}, \boldsymbol{x}, y))$
"gibbon"
99.3 % confidence

Goodfellow, Ian J., Jonathon Shlens, and Christian Szegedy (2014). "Explaining and harnessing adversarial examples." In ICLR.

Adversarial Attacks: Details

- ③ **Iterative Fast Gradient Sign Method (Iterative FGSM)**
  - **Multi-step FGSM** in an **iterative** manner.
  - A strong version of FGSM.
  - Formulation:

$$x_0^{adv} = x$$

$$x_{t+1}^{adv} = \mathrm{clip}_\epsilon \left\{ x_t^{adv} + \alpha \cdot \mathrm{sign}\left(\nabla_{x_t^{adv}} \mathcal{L}\left(f\left(x_t^{adv}; \theta \right), y_t \right) \right) \right\}$$

Clip pixel intensities to make sure the perturbation is slight.

step size

Kurakin, A., Goodfellow, I., & Bengio, S. (2016). Adversarial machine learning at scale. In ICLR.

Adversarial Attacks: Details

- ④ **One Pixel Attack**
  - Change **one pixel** in the image to fool the classifier.
  - Method: Randomly modified candidate pixels in multiple iterations and use the classification probability to choose the best one.
  - e.g.

Su, J., Vargas, D. V., & Kouichi, S. (2017). One pixel attack for fooling deep neural networks. In ICLR.

Adversarial Attacks: Details

- ⑤ **Universal Adversarial Perturbations**
  - **A perturbation** that fools a network for **"any" image**, instead of previous per-image-per-perturbation.
  - Method: **Accumulate** the perturbations computed over all the images.
  - The perturbation is shown to be **Doubly Universal** (i.e., the computed perturbation on one network also works for other networks and "any" image).

    "Transferability"

Moosavi-Dezfooli, S. M., Fawzi, A., Fawzi, O., & Frossard, P. (2017, July). Universal Adversarial Perturbations. In CVPR.

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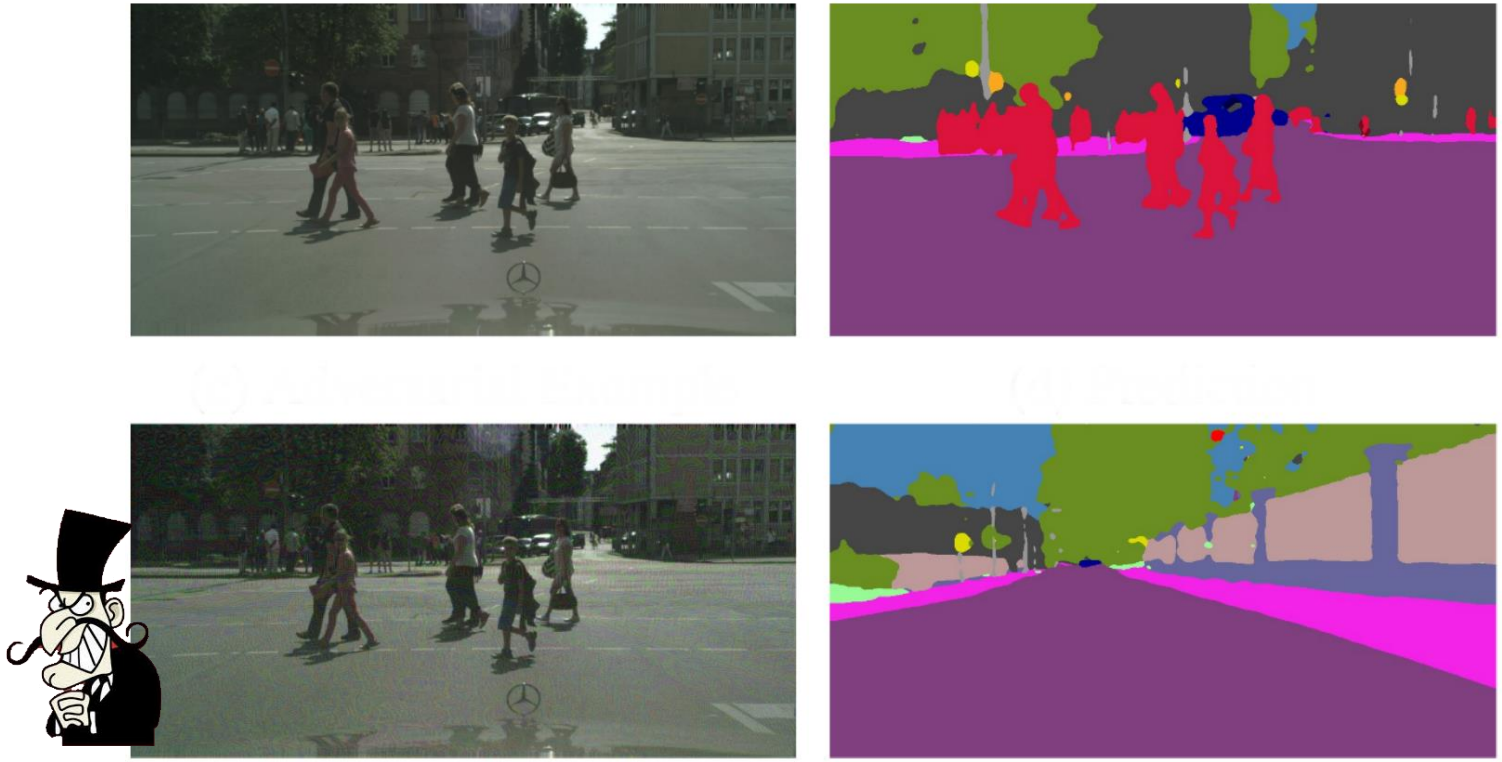
On the Robustness of Semantic Segmentation Models to Adversarial Attacks

Anurag Arnab[1] Ondrej Miksik[1,2] Philip H.S. Torr[1]

[1]University of Oxford [2]Emotech Labs

Introduction

- **First paper** to evaluate the robustness of **Semantic Segmentation** models to **Adversarial Attacks.**
  - Application: **self-driving**

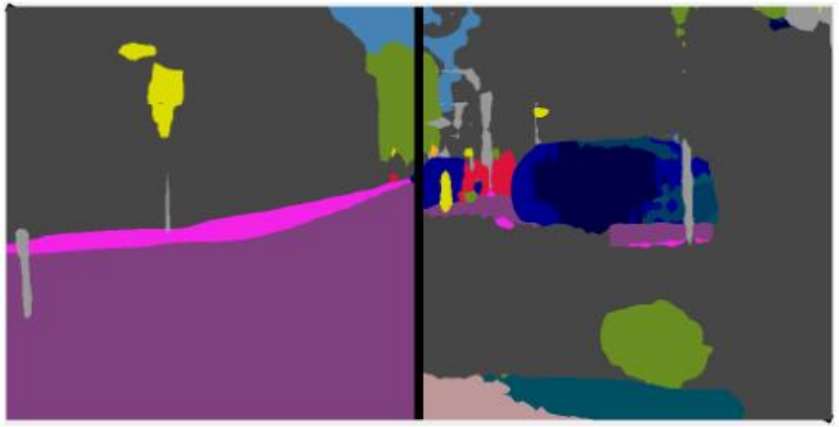


Killer

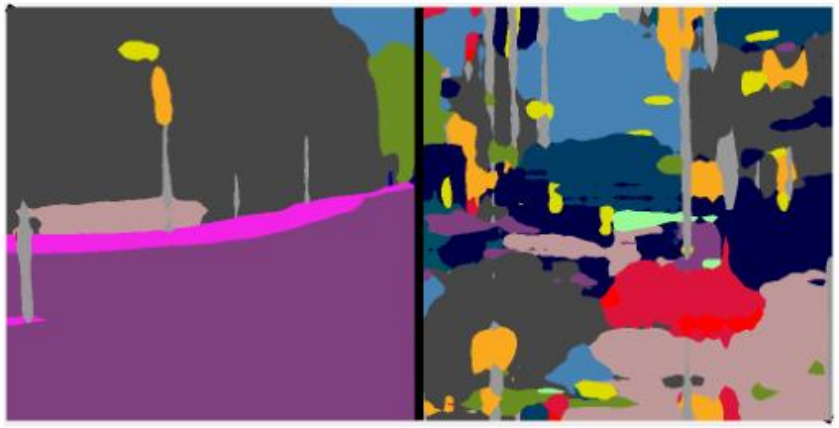
Qualitative Results

(a) Input image (perturbed half on right)

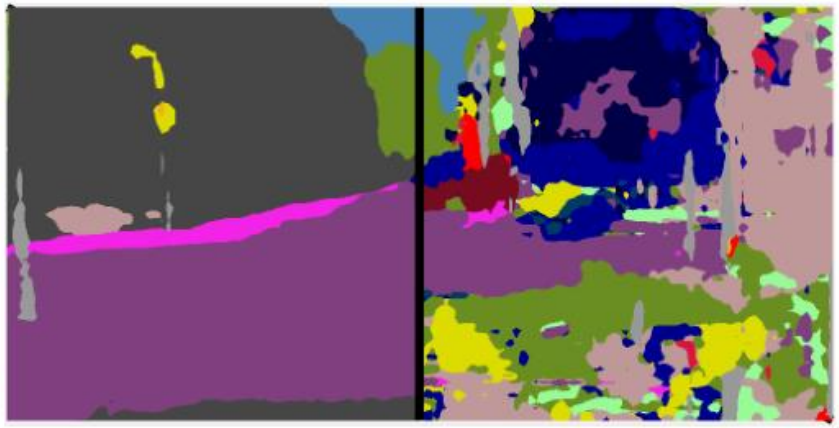
(b) Ground Truth



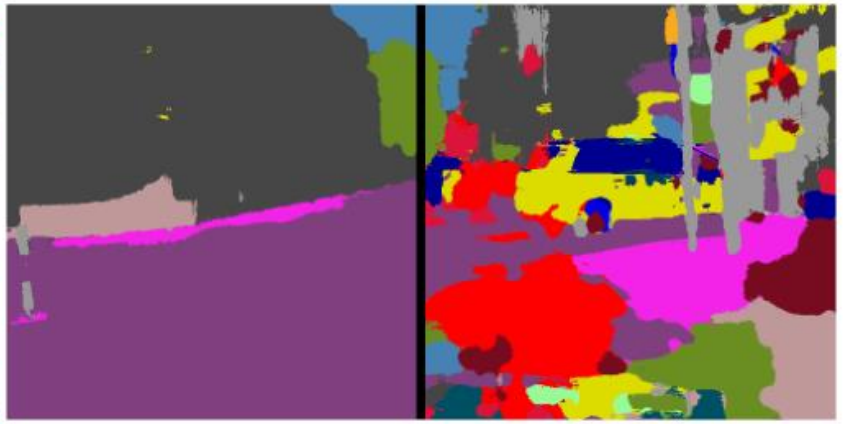
(c) PSPNet [71]



(d) DilatedNet [69]

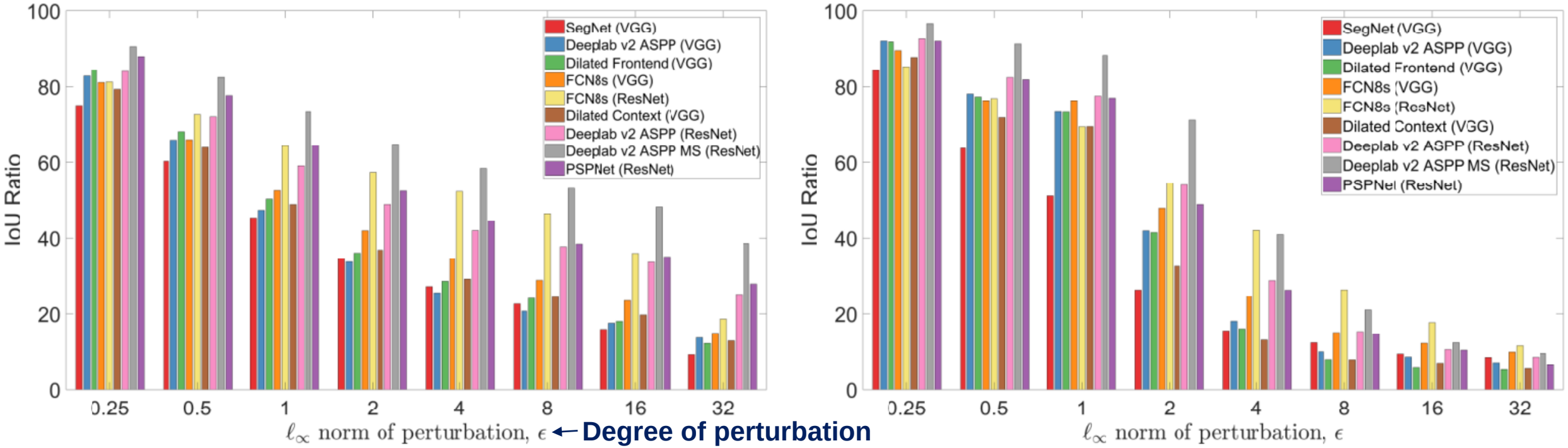


(e) ICNet [70]



(f) CRF-RNN [72]

Quantitative Results on Pascal VOC

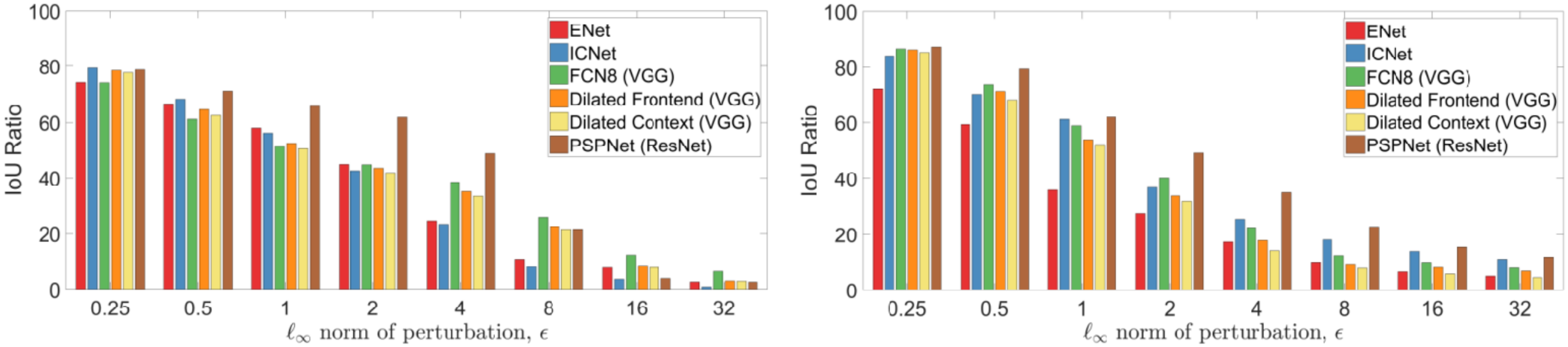


← Degree of perturbation

(a) Untargeted FGSM attack ← Different adversarial attacks (b) Targeted Iterative FGSM ll. attack

Figure 2: Adversarial robustness of state-of-the-art models on Pascal VOC. Models based on the ResNet backbone tend to be more robust. For instance, FCN8s and Deeplab v2 ASPP with a ResNet-101 backbone are more robust than with the VGG backbone. Moreover, as expected, the Iterative FGSM ll attack is more powerful at fooling networks than single-step FGSM. Models are ordered by increasing IoU on clean inputs. Results on additional attacks are in the supplementary.

Quantitative Results on the Cityscapes



(a) Untargeted FGSM attack

(b) Targeted Iterative FGSM ll. attack

Figure 3: Adversarial robustness of state-of-the-art models on the Cityscapes dataset. Contrary to Madry *et al.* [50], we observe that lightweight networks such as E-Net [58] and ICNet [70] are often about as robust as Dilated-Net [69] (341× more parameters than E-Net). Dilated-Net without its "Context" module is slightly more robust than the full network. As with the VOC dataset, ResNet (PSPNet) architectures are more robust than VGG (Dilated-Net and FCN8). Curiously, the FGSM attack is more effective than Iterative FGSM ll which computes adversarial examples from a larger search space.

Conclusions

- ① **Deeplab v2** network is significantly **more robust** than approaches which achieve better prediction scores on public benchmarks.
- ② Adversarial examples are less effective when processed at **multiple scales.**
- ③ Some widely accepted observations about robustness and model size or iterative attacks, which were made in the context of **image classification** do not transfer to **semantic segmentation** and different, real-world datasets.
- ④ Our findings will facilitate future efforts in understanding and defending against **adversarial examples** without compromising predictive accuracy.

Discussions

- Adversarial Examples widely **exist** and have many types.
- Adversarial Attack works for **Classification** models, **Generative** models, **RNN** and also **Reinforcement Learning**.
- The study of Adversarial Attack also benefits **Network Interpretation**.

Discussions

- **Pay attention** to the **area** where nobody has studied the adversarial attacks, e.g. **Zero/Few-shot Learning** (Dr. Song, 2 papers)

  Also areas where adversarial attacks are proved effective, but no **Adversarial-Aware Algorithms** are proposed, e.g. **Human Parsing** (Dr. Ye, 1 paper)
- There are somewhat fixed **patterns** for papers on adversarial attacks.

  Therefore, less time is needed to produce a paper.
- Would like to recommend some **toolboxes** for creating adversarial examples to my labmates.

CleverHans by OpenAI ★ Star 1,927

https://github.com/tensorflow/cleverhans

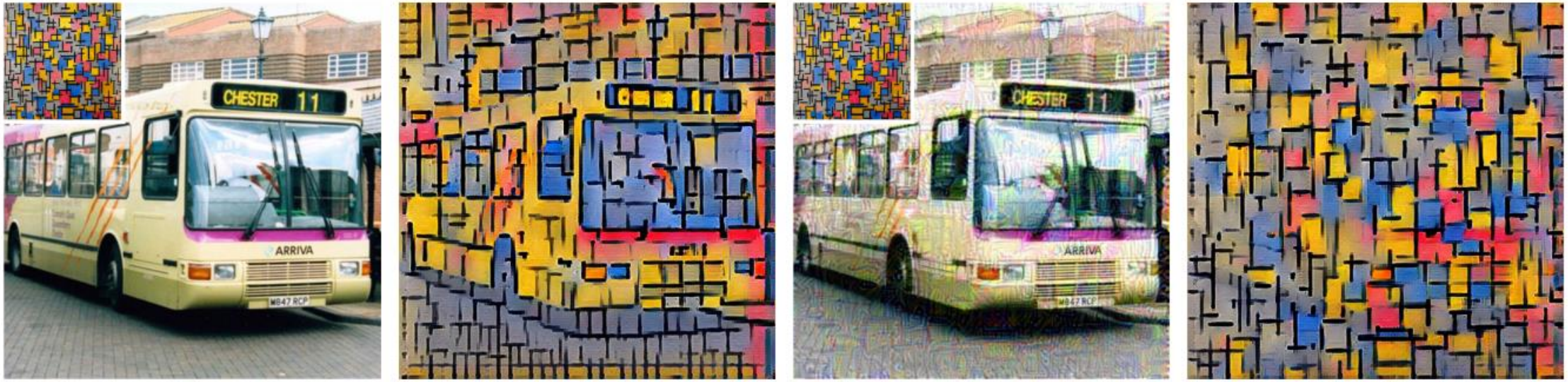
Foolbox by Bethge Lab ★ Star 492

https://github.com/bethgelab/foolbox

Discussions: Help

- **Applications** of Adversarial Examples on **Neural Style Transfer ?**

(other than network interpretation)



(a) (b) (c) (d)

Clean Image **Adversarial Example**

Thanks!